

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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TITLE

Design a Responsive College Webpage Using HTML5 and CSS3 Layout Cards

AIM

To understand and implement structural HTML5 elements combined with CSS3 styling rules (including Flexbox layout and hover animations) to build a stylized static webpage for a college website.

OBJECTIVES

- Create a structured HTML5 document with header, navigation bar, content sections, and footer.
- Apply CSS styling for background colors, font styling, padding, margins, and text alignment.
- Implement responsive layout cards using CSS Flexbox for displaying course options, facilities, and activities.
- Incorporate interactive pseudo-classes (:hover) for navigation links and card scale transitions.
- Utilize modern box sizing (box-sizing: border-box) and reset default margins.

TASK DESCRIPTION

Design a visually styled webpage titled "My College Website". The layout consists of a centered dark header section, a navigation bar with interactive links, an "About Us" section containing descriptive text, and a three-card layout displaying details on

"Courses", "Facilities", and "Activities". The cards must be styled with shadows, rounded borders, white background, and hover scaling effects. Finally, include a centered footer displaying copyright information.

ALGORITHM/PROCEDURE

1. Create an HTML document containing standard structural tags: html, head, header, nav, section, and footer.
2. Define an external or embedded stylesheet to apply global reset parameters (margin: 0, box-sizing: border-box).
3. Style the header and footer with dark backgrounds (#333), centered text, and white font colors.
4. Format navigation links with horizontal spacing and a yellow hover color effect.
5. Build a container class .cards using Flexbox layout rules to align three card modules horizontally with consistent spacing.
6. Apply visual polish to individual .card blocks using background colors, borders, drop shadows, and scale transformations on mouse hover.
7. Save and open the HTML file in a web browser to verify proper visual execution.

PROGRAM/SOURCE CODE

The following code represents the complete HTML source code and CSS styling definitions:

```
<!-- index.html -->
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My College Website</title>
  <!-- Link to external stylesheet -->
  <link rel="stylesheet" href="stylesheet.css">
</head>

<body>

  <!-- HERO HEADER: Dark background with site title and subtitle -->
  <header class="site-header">
    <div class="header-content">
      <p class="eyebrow">Learn • Grow • Succeed</p>
      <h1>My College Website</h1>
      <p>Building bright futures through education and
innovation.</p>
```

```

        </div>
    </header>

    <!-- NAVIGATION: Sticky/top bar with internal anchor links -->
    <nav class="site-nav" aria-label="Main navigation">
        <a href="#home">Home</a>
        <a href="#about">About</a>
        <a href="#courses">Courses</a>
        <a href="#contact">Contact</a>
    </nav>

    <!-- MAIN CONTENT AREA -->
    <main id="home">

        <!-- ABOUT SECTION: Introduction text -->
        <section class="intro" id="about">
            <p class="section-label">Discover your potential</p>
            <h2>Learn with purpose</h2>
            <p>Our college provides quality education and helps students
develop technical and professional skills.</p>
        </section>

        <!-- CARDS SECTION: Displays features using CSS Flexbox layout -->
        <section class="cards-section" id="courses" aria-
labelledby="opportunities-heading">
            <h2 id="opportunities-heading">What we offer</h2>

            <!-- Flex container for the 3 cards -->
            <div class="cards">

                <!-- CARD 1: Courses -->
                <article class="card">
                    <span class="card-number">01</span>
                    <h3>Courses</h3>
                    <p>Explore technical and professional courses designed
for the future.</p>
                </article>

                <!-- CARD 2: Facilities -->
                <article class="card">
                    <span class="card-number">02</span>
                    <h3>Facilities</h3>
                    <p>Use modern labs, a well-stocked library, and
student-friendly spaces.</p>
                </article>

                <!-- CARD 3: Activities -->
                <article class="card">
                    <span class="card-number">03</span>
                    <h3>Activities</h3>
                    <p>Take part in sports, cultural events, clubs, and
campus activities.</p>
            </div>
        </section>
    </main>

```

```

        </article>

    </div>
</section>
</main>

<!-- FOOTER: Copyright and closing info -->
<footer class="site-footer" id="contact">
    <p>Have questions? Visit our campus and start your journey.</p>
    <small>&copy; 2026 My College Website</small>
</footer>

</body>
</html>

```

```

/* stylesheet.css */

/* GLOBAL RESET: Ensure padding/borders don't affect widths */
* {
    box-sizing: border-box;
}

/* CSS VARIABLES: Defining the color palette for consistent theming */
:root {
    --navy: #102a43;
    --blue: #1f6f8b;
    --light-blue: #e8f3f7;
    --cream: #ffffaf2;
    --text: #243b53;
    --muted: #627d98;
}

/* BODY: Default typography and background */
body {
    margin: 0;
    font-family: "Trebuchet MS", Arial, sans-serif;
    color: var(--text);
    background-color: var(--cream);
    line-height: 1.6;
}

/* --- HEADER STYLING --- */
.site-header {
    text-align: center;
    background-color: var(--navy);
    color: white;
    padding: 70px 20px 75px;
}
.header-content {
    max-width: 760px;
}

```

```

    margin: 0 auto;
}
/* clamp() allows font size to scale fluidly with screen size */
.site-header h1 {
    margin: 8px 0;
    font-size: clamp(2rem, 5vw, 3.5rem);
    line-height: 1.1;
}
.eyebrow, .section-label {
    color: #7ed6df; /* Cyan accent color */
    font-size: 0.8rem;
    font-weight: bold;
    letter-spacing: 2px;
    text-transform: uppercase;
}

/* --- NAVIGATION STYLING --- */
.site-nav {
    text-align: center;
    background-color: white;
    border-bottom: 1px solid #d9e2ec;
    padding: 18px 20px;
}
.site-nav a {
    color: var(--navy);
    text-decoration: none;
    margin: 0 18px;
    font-weight: bold;
}
/* Hover effect for links */
.site-nav a:hover, .site-nav a:focus {
    color: var(--blue);
    text-decoration: underline;
}

/* --- MAIN SECTION LAYOUTS --- */
.intro, .cards-section {
    max-width: 1080px;
    margin: 0 auto;
    padding: 60px 24px 20px;
    text-align: center;
}
.intro h2, .cards-section h2 {
    margin: 0 0 12px;
    color: var(--navy);
    font-size: 2rem;
}

/* --- FLEXBOX CARDS CONTAINER --- */
.cards {
    display: flex;          /* Use flexbox to align side-by-side */
    flex-wrap: wrap;       /* Allow wrapping on small screens */

```

```

        gap: 20px;                /* Space between cards */
        justify-content: center;
        margin-top: 28px;
    }

/* --- INDIVIDUAL CARD STYLING --- */
.card {
    flex: 1 1 260px;             /* Grow and shrink, base width 260px */
    max-width: 320px;
    background-color: white;
    border: 1px solid #d9e2ec;
    border-top: 4px solid var(--blue); /* Blue accent on top */
    border-radius: 8px;
    padding: 28px 24px;
    text-align: left;
    box-shadow: 0 8px 20px rgba(16, 42, 67, 0.08); /* Soft drop shadow */
    transition: transform 0.2s ease, box-shadow 0.2s ease;
}

/* Hover Animation: Lift card and deepen shadow */
.card:hover {
    transform: translateY(-6px);
    box-shadow: 0 12px 24px rgba(16, 42, 67, 0.14);
}

.card-number {
    color: var(--blue);
    font-weight: bold;
    letter-spacing: 1px;
}

.card h3 {
    margin: 10px 0 8px;
    color: var(--navy);
    font-size: 1.35rem;
}

/* --- FOOTER STYLING --- */
.site-footer {
    text-align: center;
    background-color: var(--navy);
    color: white;
    margin-top: 70px;
    padding: 30px 20px;
}

/* --- RESPONSIVE DESIGN (MOBILE) --- */
@media (max-width: 600px) {
    .site-nav a {
        display: inline-block;
        margin: 4px 8px;
    }
    .site-header {

```

```
padding: 52px 20px 58px;
    }
}
```

CODE EXPLANATION

HTML/CSS Element	Purpose & Explanation
:root	Defines global CSS variables (custom properties) for consistent color theming throughout the document.
header.site-header	Defines the top banner section of the college webpage, formatted with a navy background and centered text.
nav.site-nav	Creates the main navigation menu using aria-labels for accessibility. Contains links with margin spacing.
nav a:hover	Applies an interactive hover state that changes text color and adds an underline when hovered.
main / section	Wraps the primary body content, categorizing the 'about' introduction and 'courses' sections.
.cards	Flexbox container that automatically wraps child elements (.card) and adds a 20px gap between them.
article.card	Individual card using 'flex: 1 1 260px' to maintain a fluid width. Features a top border and box-shadow.
.card:hover	Enhances user experience by moving the card upwards (translateY) and increasing shadow depth on hover.
clamp() / @media	Ensures responsive design by scaling fonts fluidly and adjusting navigation layouts on smaller screens (<600px).

TEST CASES AND EXECUTION DATA

Test Case	Verification	Expected Result	Observed Result	Status
TC-01	Page header styling	Centered white h1/p text on full-width navy blue header, with light-cyan	Header renders with navy blue background, light-cyan eyebrow text,	Pass

		eyebrow text.	white descriptive text, all centered.	
TC-02	Navigation links & hover	Centered navy blue links on white background bar; text changes to cyan on hover.	Links appear centered on a white bar; interactive hover state (cyan color, no underline) is functional.	Pass
TC-03	Card layout arrangement	Three stylized cards arranged horizontally with flexbox, side-by-side with proportional gaps.	Card container organizes cards fluidly across the page in a responsive flex row, with defined gaps.	Pass
TC-04	Card details styling	Cards are white with navy border, cyan card number (e.g., 01), blue top accent border, navy h3 heading, muted text, 8px border-radius, subtle shadow, and hover scale.	Cards render all specified visual details (number, shadow, border) correctly, including hover translateY effect.	Pass
TC-05	Footer rendering	Centered white description and copyright text on a full-width navy blue footer section.	Footer stays aligned at bottom with correct background color, centered white text (copy text and small copyright).	Pass

OUTPUT

The following browser output screenshots depict the execution of the 5 Test Cases based on the implemented HTML and CSS:

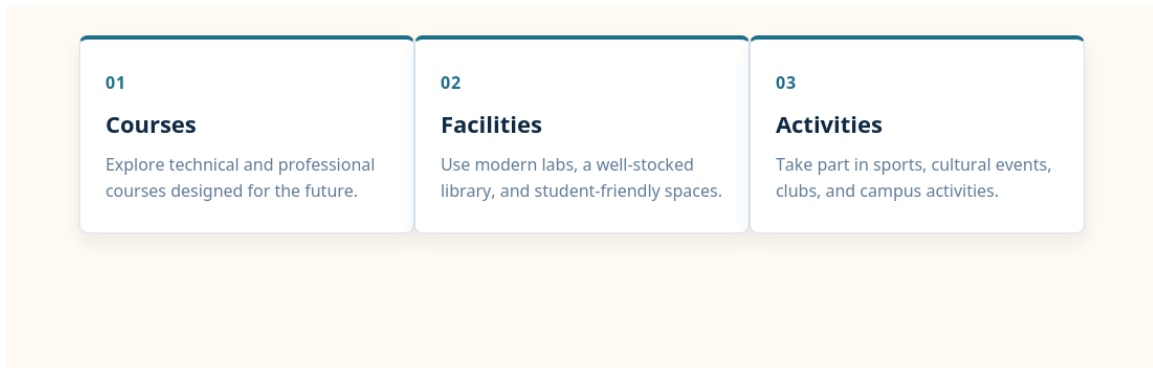
TC-01: Page header styling



TC-02: Navigation links & hover state (simulated on 'About')

Home [About](#) Courses Contact

TC-03: Card layout arrangement (Flexbox)



TC-04: Card details styling and hover effect (translateY & shadow)

01

Courses

Explore technical and professional courses designed for the future.

TC-05: Footer rendering

Have questions? Visit our campus and start your journey.

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RESULT

Thus, a styled static college webpage was successfully designed using HTML structural tags and CSS layout styles. The webpage demonstrates standard layout structure, flex card containers, box shadow styling, interactive hover transitions, and clean typography. The browser output screenshot confirms proper implementation of all design specifications.